

**UNIVERSITY OF THE WITWATERSRAND
SCHOOL OF BUSINESS SCIENCES**

**INFORMATION SYSTEMS IIA (INFO2001A)
TEST 1 – AUGUST 2025_MEMO**

DURATION: 1¾ hours (105 minutes)

MARKS: 55

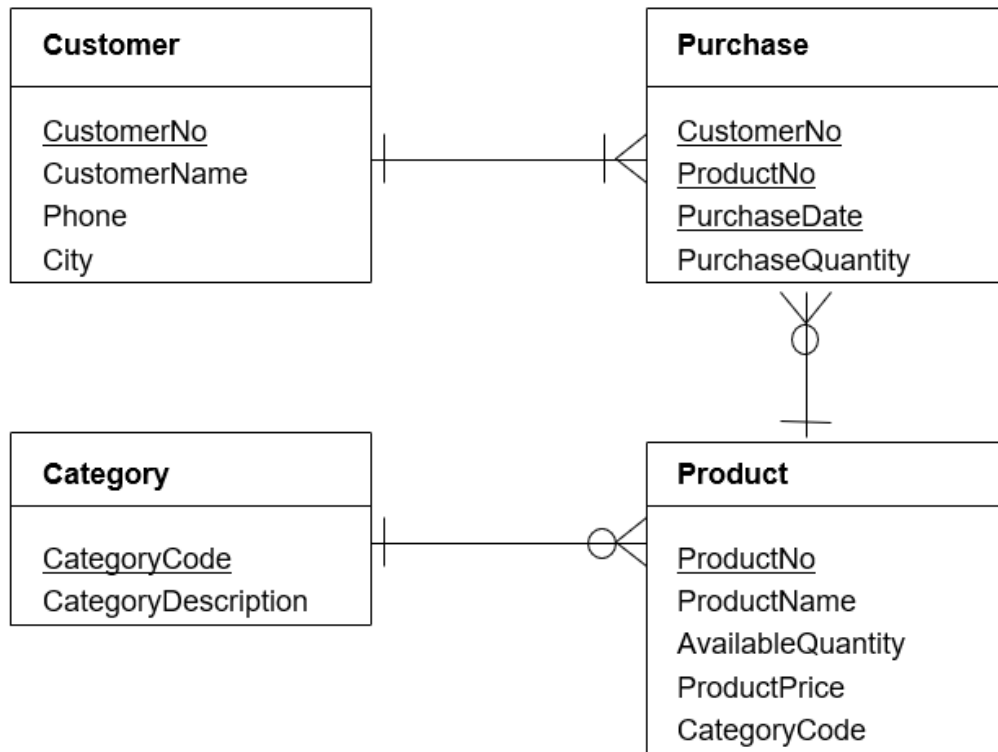
INSTRUCTIONS:

1. This paper consists of **EIGHT (8)** pages. Please ensure that you have them all.
 2. You must answer all questions and number them correctly.
 3. Sections A and B must be answered in the “**YELLOW**” booklet, while Section B must be answered in the “**GREEN**” booklet. Also put the Section(s) you are answering in the space for “**PAPER No.:**” on each booklet.
 4. You may draw diagrams in pencil.
 5. You may draw diagrams using landscape page orientation if necessary.
 6. You may **NOT** write in the right-hand margins – these are for marking purposes only.
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Section C (20 marks, 40 minutes)

Consider the following diagram. It represents a database that a small retail company uses to track the purchases that have been made by customers. Each customer may have made more than one (1) purchase, but a particular purchase would have been made by only one (1) customer. Each purchase is only for one (1) product, but a product can be purchased more than once. There are different product categories, and each product falls into one (1) category, whereas there can be many products in a category.

Please note that Questions 1 and 2 are based on the following ERD diagram, so carefully study it before attempting these questions.



The datatypes of the fields in this database are as follows:

Character: CustomerName, Phone, City, ProductName, CategoryCode, CategoryDescription

Number: CustomerNo, ProductNo, PurchaseQuantity, AvailableQuantity, ProductPrice

Date: PurchaseDate

1. Write an SQL query to find the total value of inventory on hand for each product category. The value for a single product is calculated as its available quantity multiplied by its price. The report should show the category code, the full category description, and the total inventory value for that category (heading of this column should be "Total Value"). Show only categories with a total inventory value greater than 10 000. [8]

Answer:

```

SELECT cat.CategoryCode, cat.CategoryDescription, [1]
       SUM(prod.AvailableQuantity * prod.ProductPrice) [1] AS "Total
Value" [1]
FROM Category cat, Product prod [1]
WHERE cat.CategoryCode = prod.CategoryCode [1]
GROUP BY cat.CategoryCode, cat.CategoryDescription [1]
HAVING SUM(prod.AvailableQuantity * prod.ProductPrice) 1> 10000 [1]
  
```

OR

```

SELECT cat.CategoryCode, cat.CategoryDescription, [1]
       SUM(prod.AvailableQuantity * prod.ProductPrice), [1] AS "Total
Value" [1]
  
```

```
FROM Category cat [0.5]
INNER JOIN [0.5] Product prod ON cat.CategoryCode = prod.CategoryCode [1]
GROUP BY cat.CategoryCode, cat.CategoryDescription [1]
HAVING SUM(prod.AvailableQuantity * prod.ProductPrice) [1] > 10000 [1]
```

2. The marketing team is preparing a targeted campaign for loyal customers in Cape Town. Write an SQL statement that provides a list of all products recently purchased by these customers. The report should show the product name, customer name, and purchase date. The date should be printed out in the same format as “3-June-2005” (heading of this column should be “Purchase Date”). Sort the results by the purchase date, showing the most recent purchases first. [8]

Answer:

```
SELECT p.ProductName, c.CustomerName, [1]
       FORMAT(pur.PurchaseDate, 'd-MMMM-yyyy') 1 AS Purchase Date [0.5]
FROM Customer c [0.5], Purchase pur [0.5], Product p [0.5]
WHERE c.CustomerNo = pur.CustomerNo [1]
AND pur.ProductNo = p.ProductNo [1]
AND c.City = 'Cape Town' [1]
ORDER BY pur.PurchaseDate DESC [1]
```

OR

```
SELECT p.ProductName, c.CustomerName,
       FORMAT(pur.PurchaseDate, 'd-MMMM-yyyy') AS PurchaseDate
FROM Purchase pur
INNER JOIN Customer c ON pur.CustomerNo = c.CustomerNo
INNER JOIN Product p ON pur.ProductNo = p.ProductNo
WHERE c.City = 'Cape Town'
ORDER BY pur.PurchaseDate DESC
```

3. Which SQL keyword is used to remove duplicate rows in the result set?
A. DISTINCT
 B. UNIQUE
 C. DELETE
 D. All of the above [1]
4. If one or more columns are present in a SELECT statement in addition to the SUM or AVG function, these columns need to be a part of _____. [1]
 A. HAVING clause
B. GROUP By clause
 C. ORDER By clause
 D. WHERE clause
5. Explain the difference between NULL and 0 in SQL. [2]

Answer:

NULL represents the *absence of a value*. It means the data is unknown, missing, or not applicable. [1]

0 is a *definite numeric value*. It is known and explicitly stored in the database. [1]
NULL is not equal to 0. Comparisons involving NULL typically return unknown (NULL), while
0 behaves like any other number in calculations and comparisons.

END OF TEST
